ANALYSIS AND IMPLEMENTATION OF
MULTI-OUTPUT FLYBACK DC-DC CONVERTER

Name : Mohan Sai D
Roll No : B24243
Report Period: 11 May 2026 to 10 June 2026

WORK DONE DURING THIS PERIOD

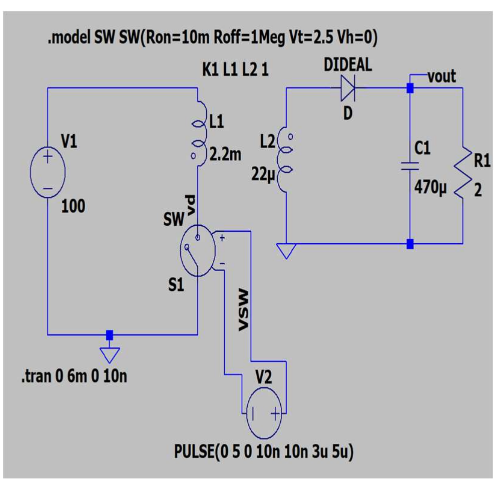
During the period from 11 May 2026 to 10 June 2026, significant progress was made on the analysis and simulation of isolated flyback DC-DC converters. The work includes studying and understanding the operating principle of flyback topology, deriving mathematical models for single-output and multi-output configurations, analyzing the effect of leakage inductance voltage spikes on the primary MOSFET switch, and designing RCD snubber circuits. All circuits were simulated using LTspice.

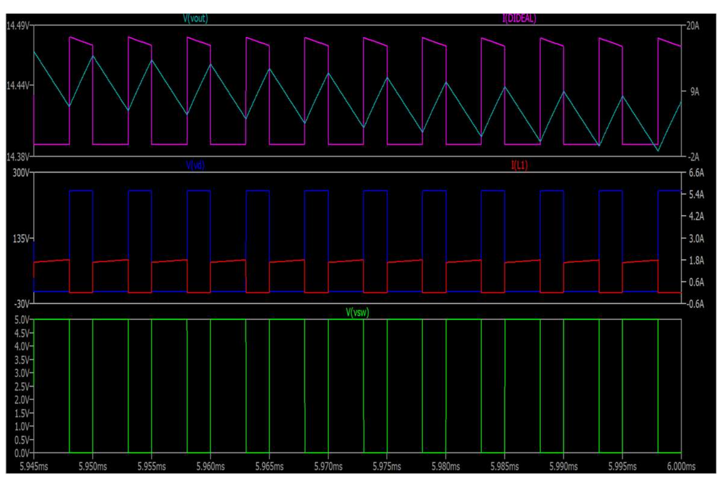
(i) Study of Flyback Converter Topology — Operating Principle and Modes

Studied the flyback converter topology in detail including its energy storage and transfer mechanism, operating principle in CCM (Continuous Conduction Mode) and DCM (Discontinuous Conduction Mode), and how it fundamentally differs from forward converters. In a flyback converter, energy is first stored in the transformer's magnetizing inductance during the ON time of the MOSFET, and then transferred to the output during the OFF time. This is fundamentally different from general transformer where energy transfer occurs simultaneously.

1.SIMULATION OF IDEAL SINGLE OUTPUT FLYBACK CONVERTER:

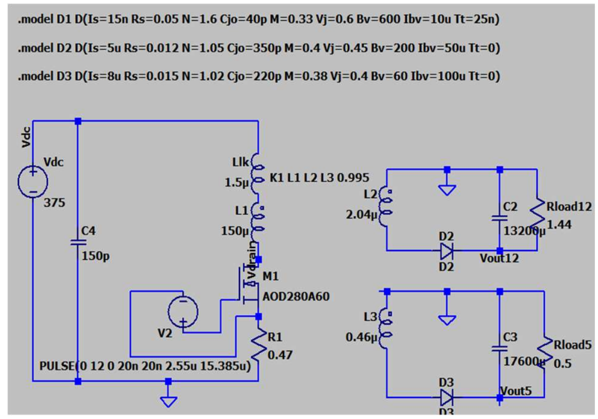
The circuit was simulated in LTspice in open-loop CCM For single output with an ideal transformer of coupling coefficient k=1 (no leakage). Below are the key waveforms : the primary current ramps up linearly during ON time, the secondary diode current ramps down during OFF time. Since the transformer is ideal here, no voltage spike appears on Vds when switch is off.

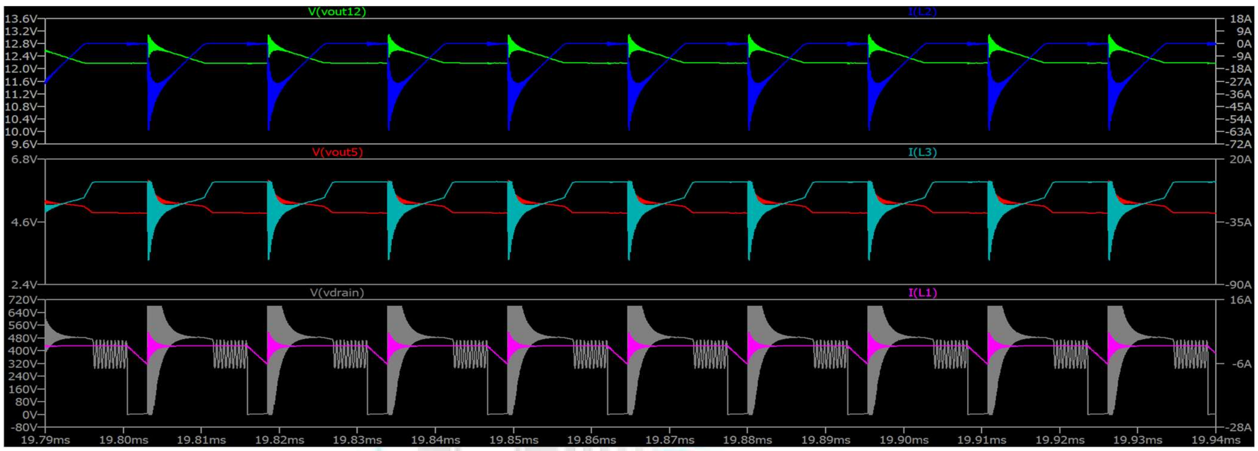




2.DESIGN AND SIMULATION OF MULTI-OUTPUT FLYBACK CONVERTER WITHOUT SNUBBER:

A multi-output flyback converter was designed with two outputs: 12V at 8.33A (100W master) and 5V at 10A (50W slave) And the total load power of 150W. The turns ratio Np:Ns1:Ns2 was calculated to satisfy volt-second balance at DC input. The circuit was first simulated without any snubber to observe the effect of leakage inductance on drain voltage and output. A non-ideal transformer model (coupling coefficient k < 1) was used so that a fraction of primary energy remains trapped as leakage inductance.

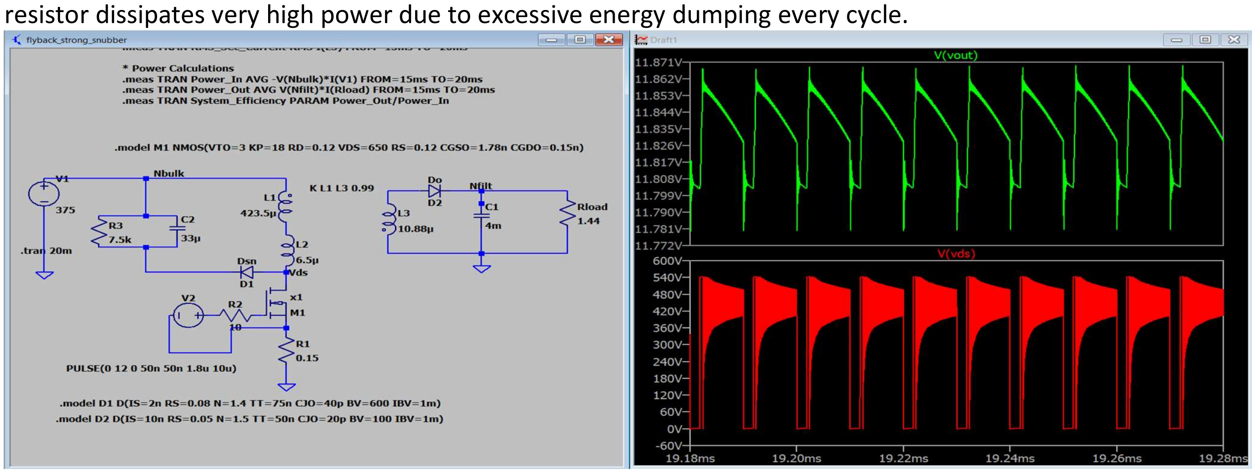




During each cycle when the MOSFET turn-off, this trapped energy produces a high unclamped voltage spike OF 680V which is very high, on the drain node, visible clearly in above figure (plot of vdrain). In a real scenario ,this spike would exceed the MOSFET's rated Vds and destroy the switch immediately. Choosing high rated Vds increases conduction losses and decreases efficiency.

3.SINGLE OUTPUT FLYBACK WITH STRONG RCD SNUBBER:

To test what happens when a very strong RCD snubber is used, a low R3 (snubber resistor) and large C2 (snubber capacitance) was intentionally applied. From the below figure, three effects were clearly observed — the spike is fully suppressed but heavy ringing appears on Vds after clamping, the snubber resistor dissipates very high power due to excessive energy dumping every cycle.

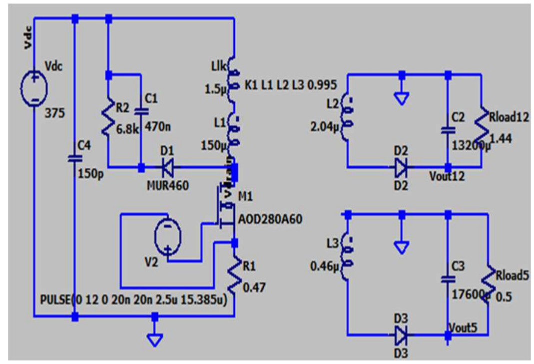


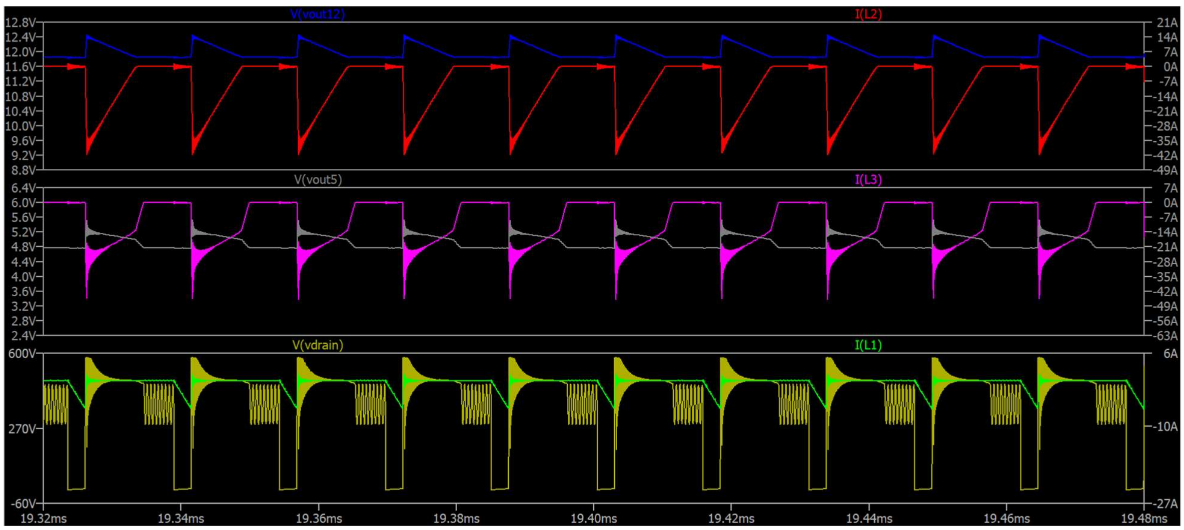
Vds waveform showing hard-clamped spike.

The output voltage dropped below 12V because the strong clamping pulls energy away before it fully transfers to the secondary. This confirmed that an over-designed snubber causes more problems than it solves, and a properly calculated snubber is necessary**.**

4.MULTI-OUTPUT FLYBACK CONVERTER WITH PROPERLY DESIGNED RCD SNUBBER:.

A properly designed snubber was then applied to the full multi-output converter. The clamp voltage set the Vds peak to 580 V, giving some margin to Vds rated voltage. R2 (snubber resistance) was selected to dissipate exactly the leakage energy per cycle without over-clamping. Below figure shows the complete schematic and waveforms — Vds is cleanly clamped, 12V and 5V outputs are stable, and the primary current ramp is undistorted. This confirms the converter is safe for testing. Comparison of Mathematical calculated values and simulated values is below.





Comparision of mathematical calculated values with simulated values:

Parameter	Mathematical calculated value	Simulation Value from Ltspice
Output Voltage (12V)	12 V	11.966 V
Output Voltage (5V)	5 V	4.933 V
Output Current (12 V)	8.333 A	8.310 A
Output Current (5 V)	10 A	9.866 A
Output Power (12 V)	100 W	99.4 W
Output Power (5 V)	50 W	48.73 W
Total Output Power	150 W	148.19 W
Input Power	187.5 W	189.52 W
Peak Primary Current	6.38 A	6.24 A
RMS Primary Current	1.5 A	1.486 A

Maximum MOSFET Vds	480 V	580.830 V
Efficiency	80.00 %	78.19 %

5.CONCLUSION:

During this reporting period, the theoretical analysis and open-loop simulations for flyback DC-DC converters were successfully completed. Mathematical models for single-output and multi-output flyback converters were derived and verified in LTspice. The effect of leakage inductance on primary switch voltage stress was observed and verified , and an RCD snubber was designed and validated.